Title: Traffic Light Control System Using Sequential Timer Logic.

Objectives:

- To understand the operation of the Timer On-Delay (TON) instruction in Rockwell CCW.
- To implement a cascading timer sequence to control multiple outputs in a predetermined time order.

System Description:

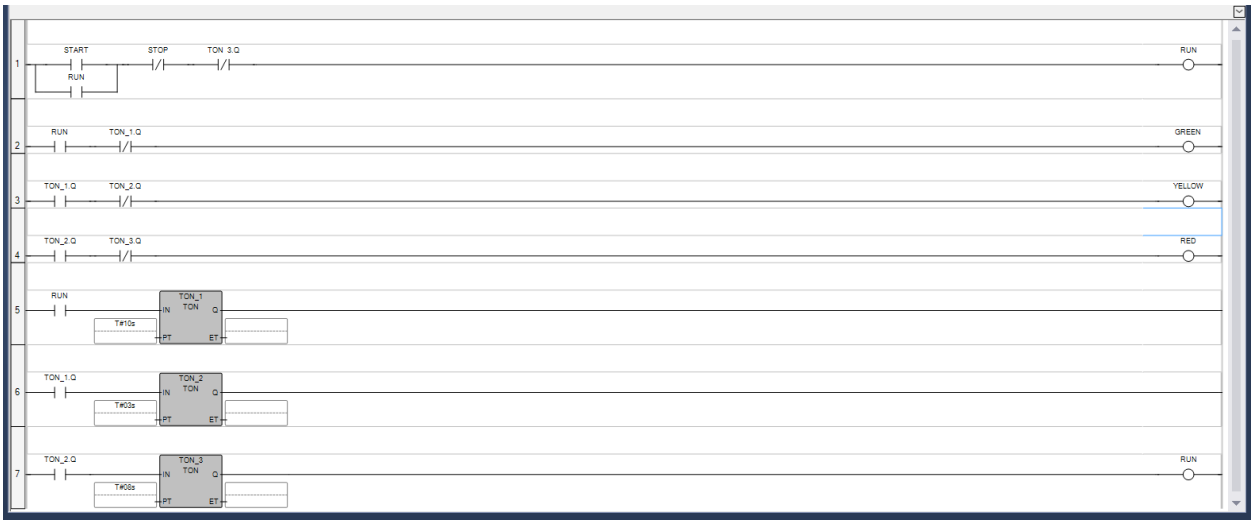
This project simulates a standard 3-light traffic intersection control system. The system operates in a continuous, repeating sequence:

GREEN Light: Illuminates for 10 seconds to allow traffic flow.

YELLOW Light: Illuminates for 3 seconds to warn of an impending red light.

RED Light: Illuminates for 8 seconds to stop traffic, before the cycle loops back to Green.

The total cycle time is 21 seconds (10s + 3s + 8s). The system is initiated by a START button and can be halted at any time by a STOP button.



The Variables of the systems are:

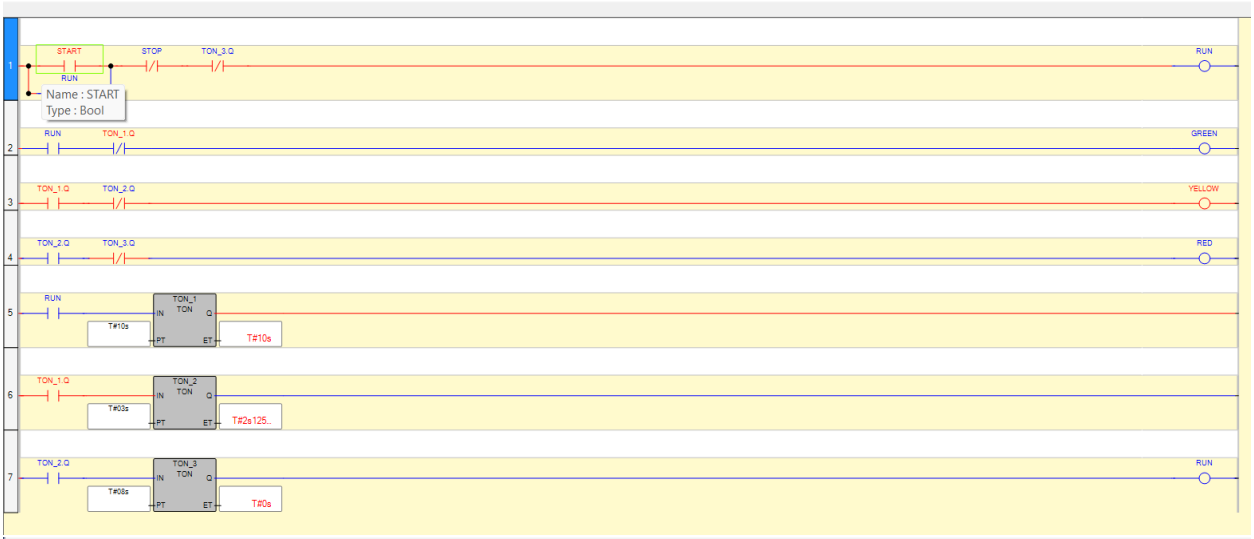
	Name	Alias	Data Type	Dimension	Project Val	Initial Value	Comment	String Size
▶	TON_1		TON		...	...		
	TON_2		TON		...	...		
	TON_3		TON		...	...		
*								

Output:


1
START
RUN
STOP
TON_3.Q
RUN
2
RUN
TON_1.Q
GREEN
3
TON_1.Q
TON_2.Q
YELLOW
4
TON_2.Q
TON_3.Q
RED
5
RUN
TON_1
TON
IN
PT
Q
ET
T#10s
T#0s
6
TON_1.Q
TON_2
TON
T#03s
T#0s
7
TON_2.Q
TON_3
TON
T#08s
T#0s
RUN


1
START
RUN
STOP
TON_3.Q
RUN
2
RUN
TON_1.Q
GREEN
3
TON_1.Q
TON_2.Q
YELLOW
4
TON_2.Q
TON_3.Q
RED
5
RUN
TON_1
TON
IN
PT
Q
ET
T#10s
T#5s813...
6
TON_1.Q
TON_2
TON
T#03s
T#0s
7
TON_2.Q
TON_3
TON
T#08s
T#0s
RUN


1
START
RUN
STOP
TON_3.Q
RUN
2
RUN
TON_1.Q
GREEN
3
TON_1.Q
TON_2.Q
YELLOW
4
TON_2.Q
TON_3.Q
RED
5
RUN
TON_1
TON
IN
PT
Q
ET
T#10s
T#10s
6
TON_1.Q
TON_2
TON
T#03s
T#3s
7
TON_2.Q
TON_3
TON
T#08s
T#3s781...
RUN


START
STOP
TON_3.Q
RUN
Name : START
Type : Bool
RUN
TON_1.Q
GREEN
TON_1.Q
TON_2.Q
YELLOW
TON_2.Q
TON_3.Q
RED
RUN
TON_1
TON
IN
Q
T#10s
PT
ET
T#10s
TON_1.Q
TON_2
TON
T#03s
T#2s125..
TON_2.Q
TON_3
TON
T#08s
T#0s
RUN


START
STOP
TON_3.Q
RUN
RUN
TON_1.Q
GREEN
TON_1.Q
TON_2.Q
YELLOW
TON_2.Q
TON_3.Q
RED
RUN
TON_1
TON
T#10s
T#8s172..
TON_1.Q
TON_2
TON
T#03s
T#0s
TON_2.Q
TON_3
TON
T#08s
T#0s
RUN


START
STOP
TON_3.Q
RUN
RUN
TON_1.Q
GREEN
TON_1.Q
TON_2.Q
YELLOW
TON_2.Q
TON_3.Q
RED
RUN
TON_1
TON
T#10s
T#0s
TON_1.Q
TON_2
TON
T#03s
T#0s
TON_2.Q
TON_3
TON
T#08s
T#0s
RUN

Discussion:

We first start the system using the “START” push button, which turns on the “RUN” bit. Now the green light turns on and Timer 1 (TON_1) starts counting to 10 seconds. When it reaches 10s, the green light turns off, the yellow light turns on, and Timer 2 (TON_2) starts counting to 3 seconds. After 3s, the yellow light turns off, the red light turns on, and Timer 3 (TON_3) starts counting to 8 seconds. Once 8s is reached, the timers automatically reset, the red light turns off, and the green light turns on again to repeat the cycle. The whole system can be turned off using the “STOP” push button.

Conclusion:

This lab successfully demonstrated the design and simulation of a time-dependent sequential traffic light control system using Rockwell CCW. The experiment reinforced the practical application of fundamental PLC programming concepts, specifically cascading Timer On-Delay (TON) instructions and continuous sequential loop logic.